

```
In [1]: import os
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
import tensorflow as tf
from tensorflow.keras import layers, models, callbacks
from helpers import *
np.random.seed(42)
tf.random.set_seed(42)
```

```
In [2]: DATASET_DIR = "../planetsdataset"
```

```
In [3]: IMG_SIZE = (64, 64)
BATCH_SIZE = 32
```

```
In [4]: train_dataset, valid_dataset, test_dataset, class_names = load_datasets(DATASET_DIR)
num_classes = len(class_names)
```

Found 2430 files belonging to 12 classes.

Found 666 files belonging to 12 classes.

Found 341 files belonging to 12 classes.

```
In [5]: num_classes = len(class_names)
#3 sets of convolution with pooling, each more complex
model = models.Sequential([
    layers.Input(shape=(IMG_SIZE[0], IMG_SIZE[1], 3)),

    layers.Rescaling(1.0 / 255),

    layers.Conv2D(32, (3, 3), activation="relu", padding="same"),
    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(64, (3, 3), activation="relu", padding="same"),
    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(128, (3, 3), activation="relu", padding="same"),
    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),
    layers.Dense(128, activation="relu"),
    layers.Dropout(0.3),
    layers.Dense(num_classes, activation="softmax")
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 64, 64, 3)	0
conv2d (Conv2D)	(None, 64, 64, 32)	896
max_pooling2d (MaxPooling2D)	(None, 32, 32, 32)	0
conv2d_1 (Conv2D)	(None, 32, 32, 64)	18496
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 64)	0
conv2d_2 (Conv2D)	(None, 16, 16, 128)	73856
max_pooling2d_2 (MaxPooling2D)	(None, 8, 8, 128)	0
flatten (Flatten)	(None, 8192)	0
dense (Dense)	(None, 128)	1048704
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 12)	1548
Total params: 1143500 (4.36 MB)		
Trainable params: 1143500 (4.36 MB)		
Non-trainable params: 0 (0.00 Byte)		

```
In [6]: early_stopping = callbacks.EarlyStopping(
        monitor="val_loss",
        patience=5,
        restore_best_weights=True
    )
```

```
In [7]: history = model.fit(
        train_dataset,
        validation_data=valid_dataset,
        epochs=20,
        callbacks=[early_stopping]
    )
```

Epoch 1/20

54/76 [=====>.....] - ETA: 2s - loss: 1.6388 - accuracy: 0.4369

2026-04-19 21:00:45.806952: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 134ms/step - loss: 1.4657 - accuracy: 0.4893 - val_loss: 0.7675 - val_accuracy: 0.7703

Epoch 2/20

54/76 [=====>.....] - ETA: 3s - loss: 0.7453 - accuracy: 0.7743

2026-04-19 21:00:57.719692: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 13s 153ms/step - loss: 0.7124 - accuracy: 0.7864 - val_loss: 0.5521 - val_accuracy: 0.8168

Epoch 3/20

54/76 [=====>.....] - ETA: 2s - loss: 0.5416 - accuracy: 0.8333

2026-04-19 21:01:09.465475: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 132ms/step - loss: 0.5146 - accuracy: 0.8379 - val_loss: 0.4369 - val_accuracy: 0.8769

Epoch 4/20

54/76 [=====>.....] - ETA: 2s - loss: 0.3869 - accuracy: 0.8808

2026-04-19 21:01:19.964948: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 10s 125ms/step - loss: 0.3751 - accuracy: 0.8835 - val_loss: 0.4344 - val_accuracy: 0.8829

Epoch 5/20

54/76 [=====>.....] - ETA: 4s - loss: 0.3495 - accuracy: 0.8866

2026-04-19 21:01:34.683532: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 20s 257ms/step - loss: 0.3538 - accuracy: 0.8860 - val_loss: 0.4109 - val_accuracy: 0.8664

Epoch 6/20

54/76 [=====>.....] - ETA: 8s - loss: 0.2667 - accuracy: 0.9213

2026-04-19 21:02:05.176206: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 33s 413ms/step - loss: 0.2559 - accuracy: 0.9276 - val_loss: 0.3532 - val_accuracy: 0.8919

Epoch 7/20

53/76 [=====>.....] - ETA: 9s - loss: 0.2010 - accuracy: 0.9381

2026-04-19 21:02:39.605518: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 30s 357ms/step - loss: 0.2032 - accuracy: 0.9379 - val_loss: 0.3880 - val_accuracy: 0.8979

Epoch 8/20

53/76 [=====>.....] - ETA: 6s - loss: 0.1990 - accuracy: 0.9393

2026-04-19 21:03:03.420193: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 26s 305ms/step - loss: 0.1917 - accuracy: 0.9412 - val_loss: 0.3840 - val_accuracy: 0.9054

Epoch 9/20

54/76 [=====>.....] - ETA: 5s - loss: 0.1458 - accuracy: 0.9525

2026-04-19 21:03:25.842589: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 20s 235ms/step - loss: 0.1409 - accuracy: 0.9551 - val_loss: 0.3533 - val_accuracy: 0.9249

Epoch 10/20

54/76 [=====>.....] - ETA: 3s - loss: 0.1295 - accuracy: 0.9583

2026-04-19 21:03:41.522335: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 15s 184ms/step - loss: 0.1318 - accuracy: 0.9551 - val_loss: 0.3683 - val_accuracy: 0.9099

Epoch 11/20

54/76 [=====>.....] - ETA: 4s - loss: 0.1201 - accuracy: 0.9583

2026-04-19 21:03:57.063514: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 17s 204ms/step - loss: 0.1160 - accuracy: 0.9597 - val_loss: 0.5947 - val_accuracy: 0.8859

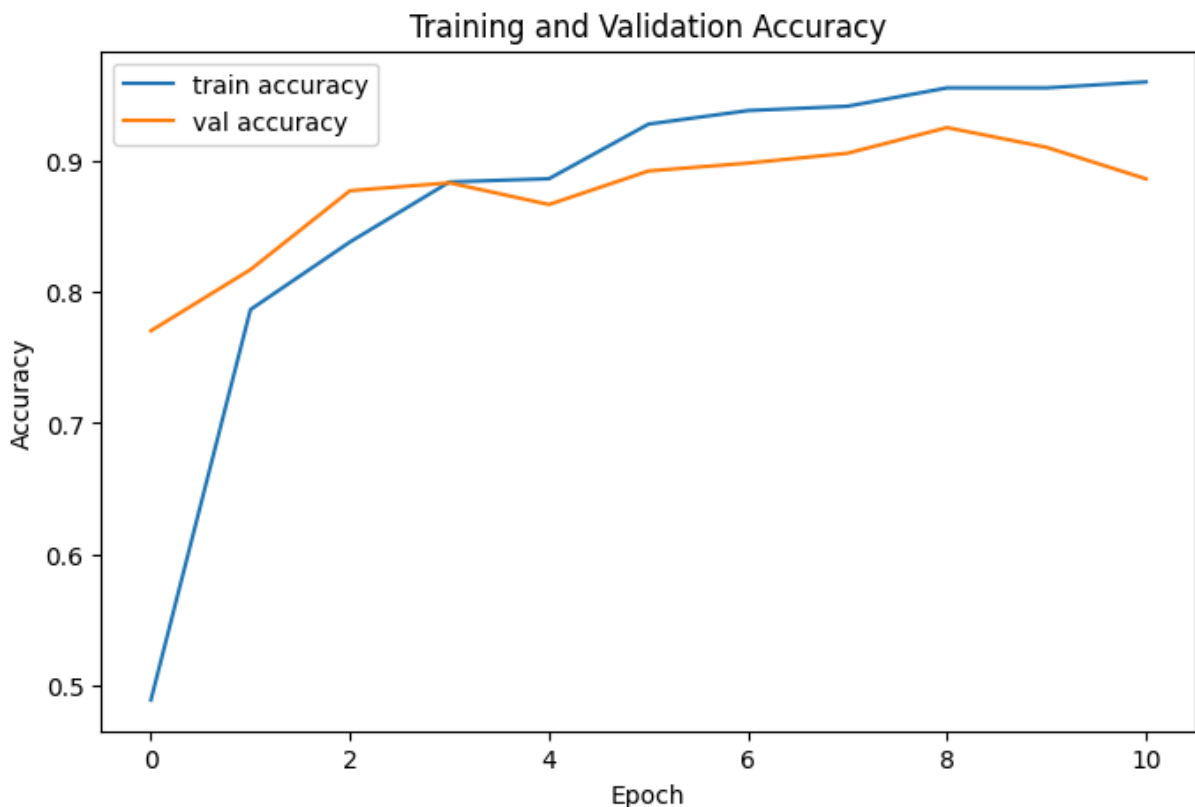
```
In [8]: test_loss, test_acc = model.evaluate(test_dataset)
print("Test loss:", test_loss)
print("Test accuracy:", test_acc)
```

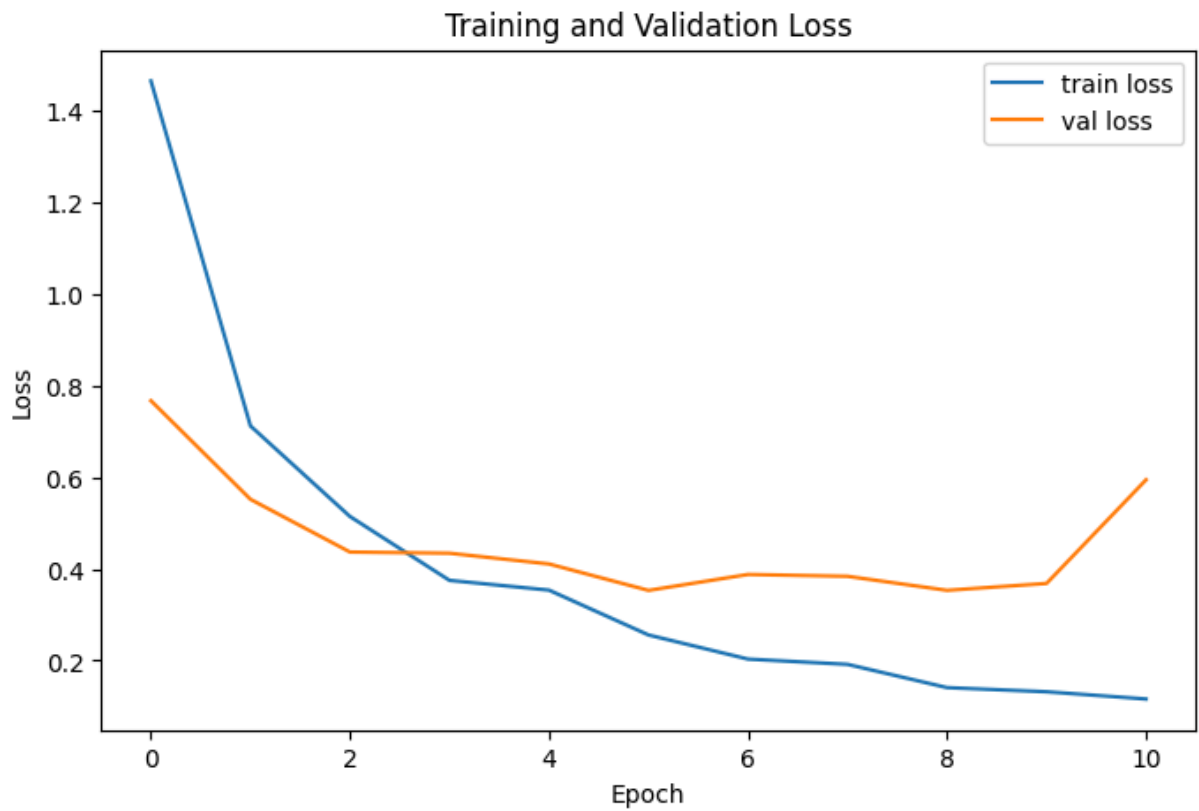
11/11 [=====] - 2s 46ms/step - loss: 0.2792 - accuracy: 0.9150

Test loss: 0.2792418897151947

Test accuracy: 0.9149560332298279

```
In [9]: plot_history(history)
```

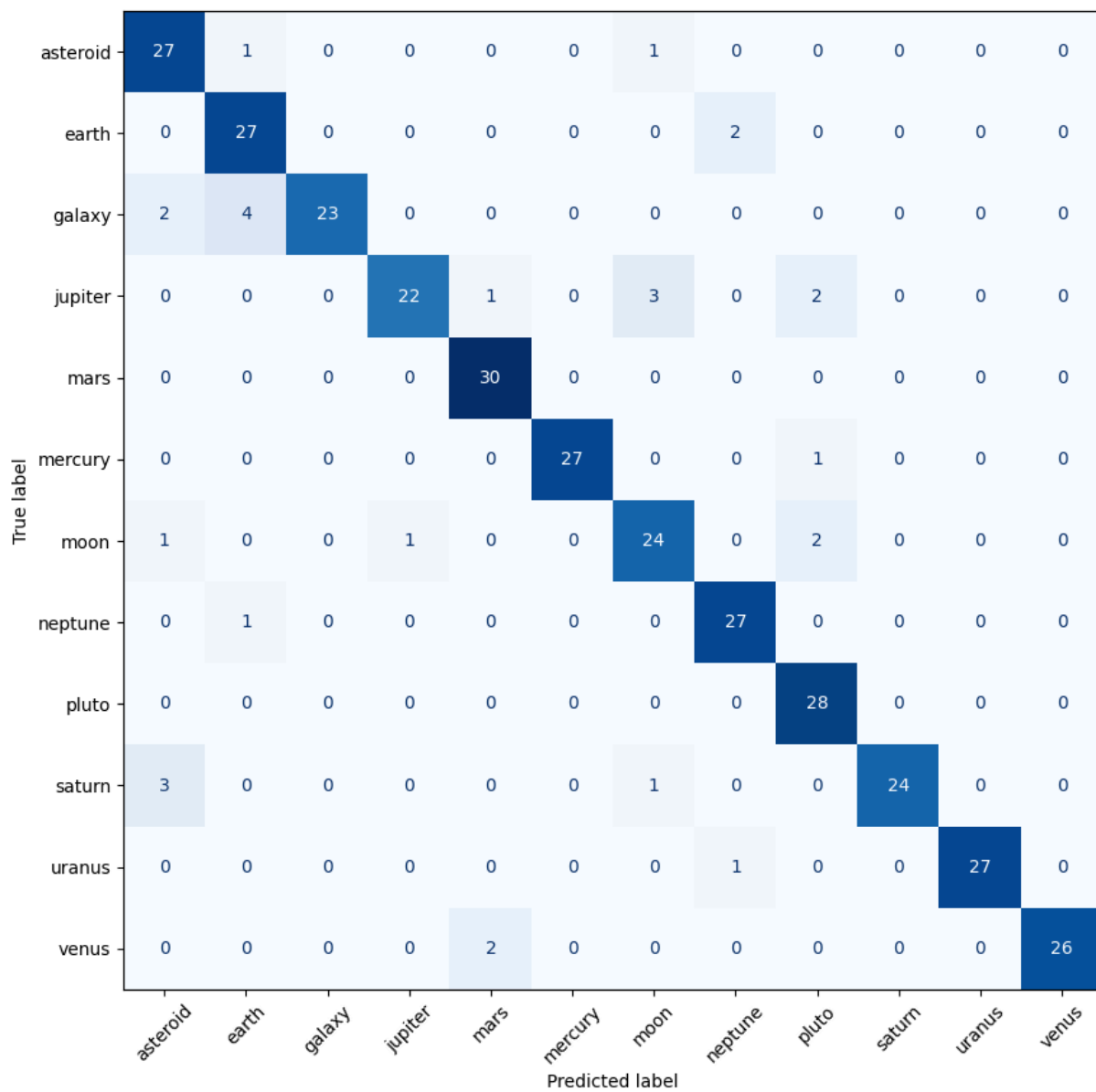




```
In [10]: y_pred_probs = model.predict(test_dataset)
y_pred = np.argmax(y_pred_probs, axis=1)
y_true = np.concatenate([y for x, y in test_dataset], axis=0)
cm = confusion_matrix(y_true, y_pred)
```

11/11 [=====] - 1s 39ms/step

```
In [11]: fig, ax = plt.subplots(figsize=(10, 10))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_name)
disp.plot(ax=ax, xticks_rotation=45, cmap="Blues", colorbar=False)
plt.show()
```



In []: